



Integrating interventional pulmonology and thoracic surgery: a multidisciplinary approach to advanced pulmonary care

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Abstract: Over the past two decades, significant advances in minimally invasive diagnostics and therapeutics have catalyzed subspecialization in pulmonary and thoracic disciplines. Interventional pulmonology (IP) has expanded the bronchoscopic management of complex airway disease, emphysema, lung cancer, and pleural pathology, while thoracic surgery (TS) continues to provide definitive operative therapies. As the diagnostic and therapeutic boundaries between these fields overlap, a coordinated multidisciplinary team (MDT) structure is warranted to optimize patient selection, reduce delays to efficient care, and offer a reasonable escalation pathway from minimally invasive to surgical approaches. At Beth Israel Deaconess Medical Center, the Chest Disease Center integrates IP and TS within a unified divisional structure supported by joint clinics, shared referral pathways, and routine multidisciplinary conferences involving radiology, oncology, and other specialties as needed. In this descriptive review, we present an institutional blueprint for integrated care using four representative disease domains: expiratory central airway collapse (ECAC), emphysema requiring lung volume reduction (LVR) strategies, early-stage lung cancer and peripheral pulmonary nodules (PPNs), and pleural disease including complicated pleural infection (CPI) and persistent air leak (PAL). Amongst those clinical entities, early dual-specialty evaluation enables timely diagnosis, individualized treatment planning, and streamlined transitions between bronchoscopic and operative options. Additionally, this framework also facilitates clinical research integration, including prospective trials embedded within routine workflows. We propose that a unified IP-TS MDT model enhances coordination, preserves continuity, and improves the efficiency of complex pulmonary care delivery, offering a practical template for adoption by other institutions seeking to align procedural innovation with patient-centered outcomes.

Keywords: Interventional pulmonology (IP); thoracic surgery (TS); multidisciplinary team (MDT); lung cancer; emphysema

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Introduction

In the past few decades, significant medical advancements have led to an extraordinary degree of medical specialization. As pathophysiological mechanisms of diseases become better understood and treatment options available to physicians become more advanced, there has been a trend towards increased sectorization of medical and surgical specialties into advanced areas of practice. This has been especially true in procedural fields, whereby the advent of minimally invasive and endoscopic platforms to provide higher-quality patient care has led to the development of new specialized niches and training opportunities. This new realm of medical care remains in constant flux due to rapid changes in the technology available (1). This can create unique challenges when it comes to collaborative diagnostic and therapeutic decision-making between different specialties. This becomes paramount amongst specialists who treat pathologies within the same organ system.

The emerging field of interventional pulmonology (IP) perfectly encapsulates this process. Advances in bronchoscopic technology and improvements in the availability of bronchoscopic tools and techniques have allowed this offshoot of pulmonary medicine to establish an essential niche in the management of pulmonary disease and thoracic malignancies. There have been many therapeutic advancements that led to the expansion of this new subspecialty including: (I) the management of central airway disease via rigid bronchoscopy; (II) the development of endobronchial laser therapy, electrocautery, cryotherapy, and endobronchial or tracheal stents (2); (III) the treatment of severe chronic obstructive pulmonary disease (COPD) with endobronchial valves (EBVs) (3); and (IV) the establishment of minimally invasive approaches such as endobronchial ultrasound (EBUS), electromagnetic navigation bronchoscopy (ENB), and robotic bronchoscopy to diagnose and treat pleural disease as well as thoracic malignancies (4).

However, despite its broad reach, IP remains a niche subspecialty that depends largely on collaboration amongst general pulmonologists, diagnostic and interventional radiologists, oncologists, and thoracic surgeons (2). For this reason, the concept of the multidisciplinary team (MDT) is essential. The MDT approach emerged in oncology in the mid-1980s with the goal of harnessing therapeutic collaborations from multiple oncologic specialties to combine chemotherapy, radiotherapy, and/or surgery (5). Oncologic guidelines now stress its importance in enhancing clinical outcomes, and it has been adopted in most developed

countries as the gold standard in cancer care delivery (6). There is a wealth of literature describing the factors that contribute to an effective MDT workflow, such as team composition, referral protocols, and clear decision-making algorithms (7). The advantages of MDT include increased patient volumes, improved clinician knowledge and technical skills through multidisciplinary discussions, and greater chances for patients to receive coordinated and personalized care (8). Ideally, the various team members of the MDT should remain involved throughout the course of clinical care, including diagnosis, treatment, and follow-up. The MDT approach has since been adopted in a variety of clinical fields such as orthopedics, musculoskeletal medicine, wound care, and urology (9).

The multidisciplinary framework serves itself well for the modern procedural management of complex thoracic pathologies, which requires close and coordinated collaboration between a multitude of providers, including thoracic surgeons and interventional pulmonologists. The Chest Disease Center at Beth Israel Deaconess Medical Center adopted this approach to capitalize on the strengths of both specialties in order to expedite work-up, diversify treatment options, and improve outcomes for patients with complex pulmonary diseases. At The Chest Disease Center, this approach is facilitated through a hybrid structure involving weekly/monthly multidisciplinary case conferences, IP, thoracic surgery (TS), radiology, and oncology as needed. It is also portrayed via a unified departmental structure in which IP and TS are housed within the same division with adjacent clinical offices. Patients with complex chest disease are typically reviewed jointly early in the care pathway. In other cases, patients may initially be evaluated by one specialty with prompt consultation of the other when multidisciplinary input is required. This framework facilitates timely decision-making while preserving continuity of care.

This paper describes the clinical experience gained from such collaboration, using four distinct clinical entities as examples: expiratory central airway collapse (ECAC), emphysema, lung cancer, and pleural disease.

Clinical applications of the multidisciplinary IP-thoracic surgery model

ECAC

ECAC is an airway disorder defined by two pathologic entities: tracheobronchomalacia (TBM) and excessive

dynamic airway collapse (EDAC) (10-13). Both conditions present with nonspecific symptoms such as dyspnea, cough, and recurrent respiratory infections (14). While sharing a similar presentation, TBM and EDAC are characterized by distinct pathophysiology. TBM involves weakening of the cartilaginous rings supporting the anterior and lateral walls of the tracheobronchial tree leading to airway narrowing. On the other hand, EDAC is characterized by the forward displacement of the posterior membrane of the trachea during forceful expiration. Both processes lead to severe dynamic airway collapse with important physiological consequences. The clinical vagueness and non-specificity of ECAC symptoms mirror those of other common respiratory comorbidities—such as asthma and COPD—and gastroesophageal reflux disease (GERD) (15). This symptomatic overlap often delays diagnosis and complicates management.

The complex nature of ECAC warrants the integration of multiple specialties, including pulmonary medicine, IP, radiology, gastroenterology, otorhinolaryngology, and TS, to ensure efficient care coordination and delivery (15). Patients with suspected ECAC should be referred from primary care providers or general pulmonologists to a specialized airway center (i.e., the Chest Disease Center). The diagnosis of ECAC can then be ascertained following a thorough tracheobronchial assessment, which includes interviews, pulmonary function tests (PFTs), and dynamic computed tomography (CT) with inspiratory and expiratory sequences. However, confirmation must be done via an awake dynamic bronchoscopy with conscious sedation (16). An observed collapse (>90%) in the cross-sectional airway is diagnostic of severe ECAC, and these patients should be considered for definitive surgical treatment via tracheobronchoplasty (TBP) (17,18). At The Chest Disease Center, to optimize identification of patients who would be most likely to benefit from TBP, the IP service offers patients with severe ECAC a temporary 2-week airway stent trial; this provides the MDT, including the surgeon with prognostic information that becomes critical in preoperative planning and surgical decision making (Video S1) (11,19). From our experience, up to 75% of patients who undergo stent trial report significant symptomatic improvement, with 80% of those who respond positively subsequently reporting improvements in quality of life (QoL) after TBP (20). Although TBP provides relief for the majority of those suffering from severe ECAC, not all patients are suitable candidates (21). Despite significant advancements in surgical technique with the advent of minimally invasive approaches, TBP remains a complex

operation with potential for morbidity, especially in frail patients with high comorbidity burdens (22,23). Thus, a stent trial to assess surgical candidacy is paramount before proceeding with TBP. Moreover, in selected high-risk patients considered to be at prohibitive risk of complications with TBP after MDT evaluation, stents can be considered as a definitive management strategy.

This diagnostic and therapeutic framework exemplifies the strengths of adopting an MDT approach (Figure 1). Combined IP and surgical evaluation is crucial to: (I) appropriately diagnose patients; (II) identify and expedite treatment for those who are likely to benefit from surgical fixation; and (III) provide alternative management options for those deemed not suitable for surgery. Such an approach facilitates enhanced patient-centered care, minimizes delays in treatment, and eliminates duplicated work and unnecessary use of healthcare resources.

Emphysema

COPD is a highly prevalent chronic lung disease affecting nearly 300 million people worldwide (24). It is often complicated by frequent exacerbations, thereby imposing substantial burden on both healthcare systems and patients' QoL (25). Due to the heterogeneity of COPD, early identification of the different phenotypes and treatable traits is critical to ensure optimal management, either through surgical or bronchoscopic interventions (26). For instance, in patients with severe COPD-related hyperinflation, pharmacologic treatment alone has very limited efficacy in improving pulmonary function. In such circumstances, lung volume reduction (LVR) strategies should be considered to reduce hyperinflation by selectively targeting diseased lung areas. This results in decreased residual volume, increasing inspiratory capacity, elastic recoil, and expiratory airflow, which promotes improved exertional tolerance and QoL (27). LVR can be achieved via surgical or bronchoscopic intervention. Though the National Emphysema Treatment Trial (NETT) found a significantly higher 90-day mortality rate for the surgical group compared to medical treatment (7.9% vs. 1.3%) (28), data from high-volume and experienced centers showed that, in carefully selected patients, surgical LVR has low mortality rates and yields sustained functional improvement lasting up to 5 years (29,30). Unfortunately, many patients remain ineligible for surgical interventions due to advanced age and/or comorbidities. Bronchoscopic LVR (BLVR) is an alternative approach which achieves LVR through the placement of

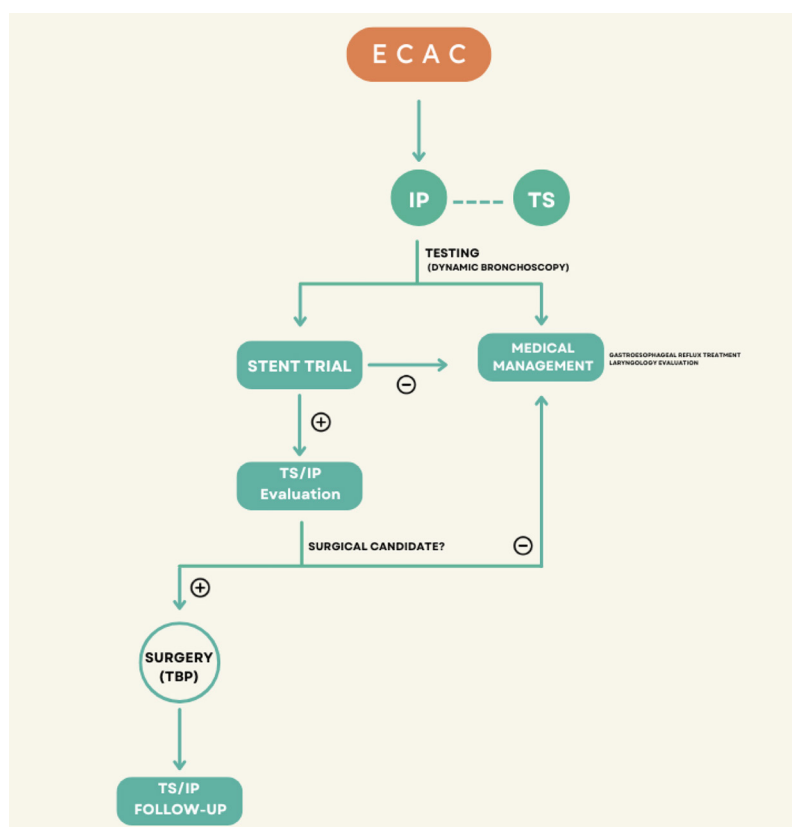


Figure 1 Multidisciplinary care pathway for the management of ECAC. Initial testing is performed by IP using dynamic bronchoscopy, with TS involvement early in the evaluation process. Patients with severe ECAC are offered a stent trial. A positive response (+) prompts joint TS/IP evaluation to determine surgical candidacy. Surgical candidates proceed to TBP, followed by joint TS/IP postoperative follow-up. A negative stent trial (-) or non-surgical candidacy leads to conservative medical management, which may include gastroesophageal reflux treatment and laryngology evaluation. This algorithm illustrates a coordinated multidisciplinary approach to optimize outcomes and avoid unnecessary interventions. The figure was created on an online platform (Canva). The editable online version can be accessed at https://www.canva.com/design/DAGIbMbqQFw/dF4uvGDcFdapwGLcCuitrw/edit?utm_content=DAGIbMbqQFw&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton. ECAC, expiratory central airway collapse; IP, interventional pulmonology; TBP, tracheobronchoplasty; TS, thoracic surgery.

one-way EBVs. These devices are designed to only allow air to flow in one direction (i.e., out of the inflated lobe), facilitating deflation of the targeted lobe (31). Recognizing their efficacy, the Global Initiative for Obstructive Lung Disease (GOLD) guidelines listed EBVs as an evidence-level A treatment option for patients with severe lung emphysema and hyperinflation (32). Compared to the surgical approach, EBVs are less invasive and reversible with evident benefits, including on cardiac function (33,34). However, successful BLVR via EBV placement requires interlobar fissures to be complete to reduce collateral ventilation, which remains a major obstacle for a subset of patients with severe hyperinflation who would otherwise

benefit from LVR (35). Estimates show that only 65% of otherwise eligible patients can undergo BLVR with EBVs because of CV (35).

Treatment of complex COPD is another prime example of how adopting an MDT approach that integrates IP and TS offers a powerful synergy that balances immediate relief with long-term management (Figure 2). An MDT framework also allows for the seamless implementation of clinical trials within the treatment workflow. For example, at The Chest Disease Center, we are conducting a randomized controlled study to compare medical management to thoracoscopic fissure completion followed by EBV placement for combined surgical-BLVR in patients with incomplete lobar

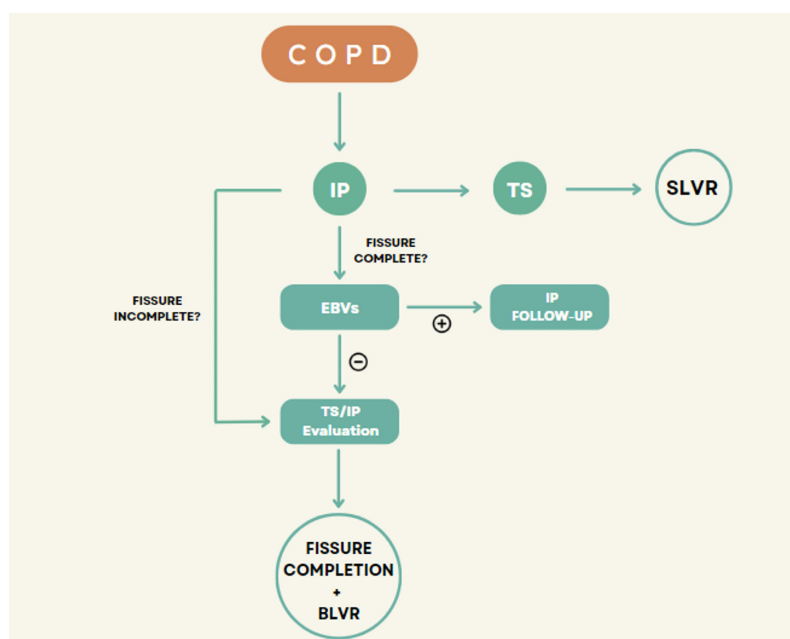


Figure 2 Institutional multidisciplinary care pathway for LVR in emphysema. This flowchart illustrates the decision-making process for LVR in patients with severe emphysema. Patients are evaluated jointly by IP and TS. If the fissure is complete, EBV) are considered. A positive response (+) to EBVs leads to IP-directed follow-up, whereas a negative response (–) prompts further evaluation by both IP and TS. For patients with incomplete fissures, a combined approach of thoracoscopic fissure completion and BLVR is considered. Patients who are optimal candidates for surgery may proceed directly to SLVR. The integration of both specialties allows for a tailored approach, balancing minimally invasive and surgical strategies to optimize patient outcomes. The figure was created on an online platform (Canva). The editable online version can be accessed at https://www.canva.com/design/DAGIbMbqQFw/dF4uvGDcFdapwGLcCuitrw/edit?utm_content=DAGIbMbqQFw&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton. BLVR, bronchoscopic lung volume reduction; COPD, chronic obstructive pulmonary disease; EBV, endobronchial valve; IP, interventional pulmonology; LVR, lung volume reduction; SLVR, surgical lung volume reduction; TS, thoracic surgery.

fissures (COMPLETE-1 trial) (Video S2) (36). Similarly, the SAVED-I study, also currently active at The Chest Disease Center, is a prospective observational study to evaluate the potential role of thoracoscopic fissure completion with pleural adhesiolysis in patients with severe emphysema who failed BLVR (37). These two trials represent yet another dimension of MDT that involves clinical research collaboration to re-shape treatment guidelines and advance standard of care. Surgical involvement in the multidisciplinary care of such patients can also allow for simultaneous resection of high-risk pulmonary nodules, which are prevalent in severe COPD and can complicate decision-making regarding target lobe in BLVR.

Lung cancer

Lung cancer is the leading cause of cancer-related mortality

worldwide responsible for over 1.8 million deaths globally (38,39). The high mortality rate is largely attributed to late diagnosis in approximately 70% of patients due to the asymptomatic nature of early disease (40). Early-stage disease is often asymptomatic, especially in the case of small peripheral pulmonary nodules (PPNs). However, implementation of low-dose CT (LDCT) screening, supported by large randomized trials such as the NLST and NELSON, has led to significant reductions in lung cancer mortality and is now recommended in multiple national guidelines (41).

In parallel, the increasing frequency of CT imaging in today's medical practice has led to a high number of incidentally found PPNs. Many of these nodules are in peripheral or difficult-to-access locations, and their evaluation requires collaboration across specialties involving oncology, radiology, TS, radiation oncology, and pulmonary

medicine specialists (39).

There have been significant advancements in IP that have dramatically improved the accuracy and safety of lung cancer diagnosis (42). Technologies such as linear EBUS have become the standard of care for mediastinal staging and tissue acquisition, with high sensitivity and specificity for nodal assessment and guided sampling of mediastinal lymph nodes (39). This not only enables histologic diagnosis, but also comprehensive molecular profiling by detecting actionable mutations, such as *EGFR*, *ALK*, and *KRAS*, which are critical for guiding targeted therapy (43,44).

Beyond diagnostics, a multidisciplinary approach also plays a crucial role in defining optimal treatment strategies. Navigational bronchoscopy, for example, can be used to aid radiation oncologists by placing fiducial markers in patients undergoing stereotactic body radiation therapy (SBRT), which enables more precise targeting of tumors (45). Similarly, thanks to ENB and virtual bronchoscopic navigation, previously inaccessible, small PPNs can now be biopsied and evaluated in real time by IP. This has tremendous implications for the surgical management of hard-to-access lesions and nodules for which histopathological analysis is required prior to resection.

Robotic-assisted bronchoscopy is another rapidly evolving tool in the field of IP that has been shown to significantly enhance the care of patients with early-stage lung cancer (46). The robotic platform allows providers to navigate along complex pathways to reach peripheral lung nodules; this has greatly increased our capability to biopsy and/or mark lesions that were previously considered inaccessible via conventional bronchoscopy. In a recent retrospective case-control study, the authors detailed the blueprint for an MDT approach in early-stage non-small cell lung cancer (NSCLC), which combines robotic-assisted bronchoscopy and surgical lobectomy in a single anesthesia, same-procedure event to reduce time between detection and definitive treatment. This approach not only largely decreased time to intervention (by an average of 36 days) but also significantly lowered complications compared to conventional biopsy (0% *vs.* 10%) (47). The Chest Disease Center has adopted this approach and completed >15 cases of combined single-procedure lesion marking followed by surgical resection (“diagnose and resect cases”) (*Figure 3*); this is particularly helpful for small (<1 cm), deeply located, or subsolid/ground-glass nodules that are unlikely to be identified visually or through palpation at the time of surgery (*Video S3*) (48). This illustrates how combining IP’s

diagnostic precision and minimally invasive techniques with TS’s curative potential ensures comprehensive and rapid care.

Pleural disease [complicated pleural infection (CPI) and persistent air leak (PAL)]

CPIs and PALs pose significant management challenges, where timely and coordinated multidisciplinary intervention is essential to optimize outcomes while balancing invasiveness with efficacy (47). For instance, CPIs affect approximately 80,000 patients each year in the United States and the United Kingdom (49). These infections are associated with high morbidity and all-cause mortality (up to 50% in intensive care unit patients), owing largely to mismanagement and delayed intervention (50). Initial management typically involves drainage and broad-spectrum antibiotic therapy, but in many cases, more aggressive interventions are indicated. Options include surgical decortication or intrapleural fibrinolytic therapy (IPFT) using tissue plasminogen activator (tPA) or deoxyribonuclease (DNase) (49).

There remains limited direct evidence comparing these two modalities, leading to variability in practice patterns. Although the Multicenter Intrapleural Sepsis Trial (MIST) II has demonstrated the efficacy of IPFT, the majority of guidelines continue to recommend surgery as the preferred approach (51). A prospective clinical trial comparing intrapleural tPA plus DNase therapy to surgical decortication in 26 patients found comparable complication rates between both groups (49). While chest tube duration was similar (5 days for IPFT *vs.* 4 days for surgery), the median hospital stay was longer for the IPFT group (11 *vs.* 5 days), though this difference was not statistically significant (49). Similarly, another multicenter study concluded that surgery resulted in shorter drainage and hospital stays. However, the fibrinolytic group required additional drain placement or surgical intervention in a significant number of cases (52).

At The Chest Disease Center, CPI management follows a coordinated pathway; all patients are jointly evaluated by IP and TS early during their admission. TS directs care in patients deemed fit for surgery, aiming for early definitive intervention. In those with prohibitive surgical risk, IPFT is performed under close surgical oversight, with rapid escalation to operative management if there is no clinical improvement. This parallel involvement avoids delays, ensures readiness for escalation, and optimizes resource utilization. The same approach is utilized for patients with

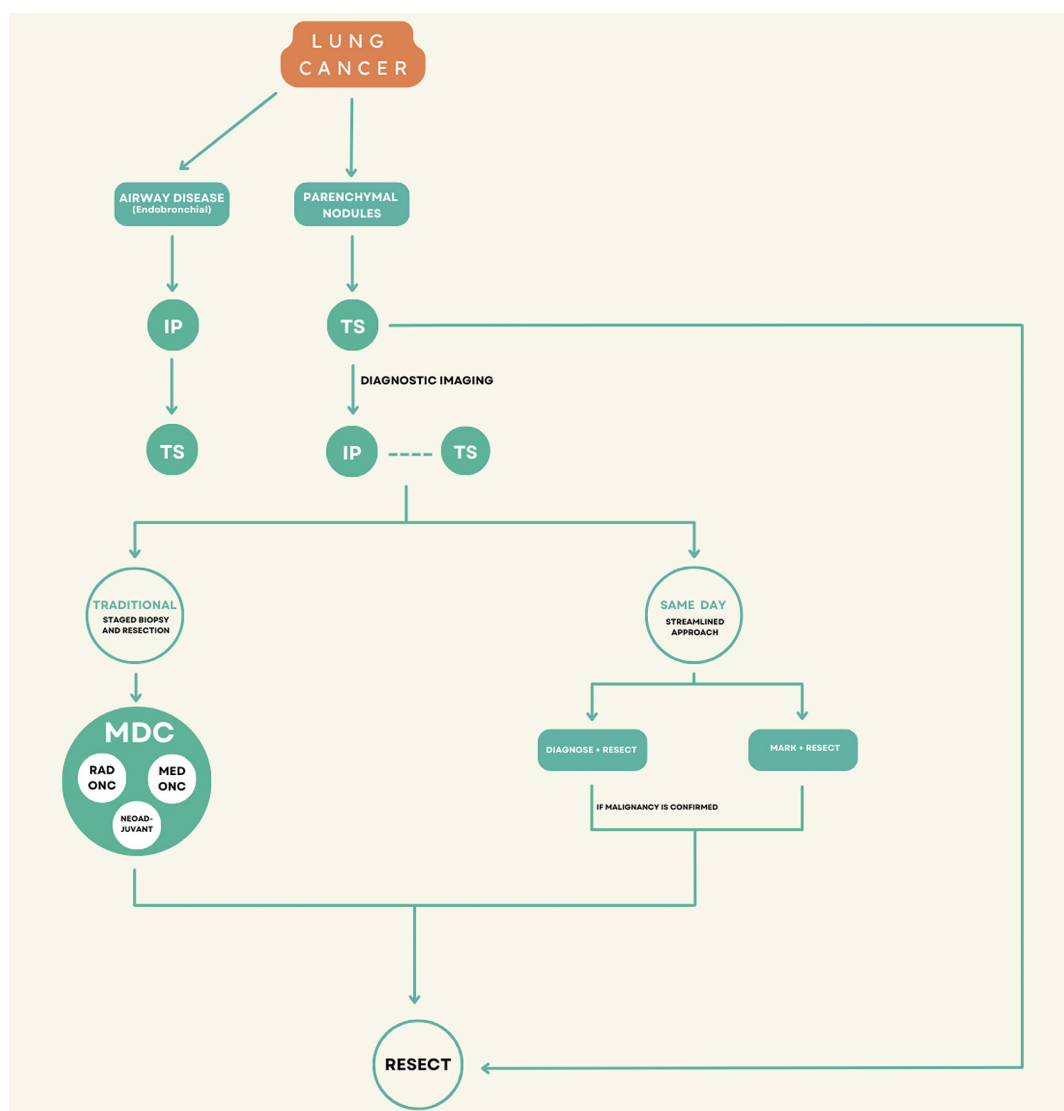


Figure 3 Lung cancer may present as either airway or parenchymal disease, guiding initial evaluation by IP or TS, respectively. Diagnostic imaging and collaborative evaluation between IP and TS determine the appropriate procedural pathway. The traditional approach involves a staged biopsy followed by MDC review—including Med Onc and Rad Onc—for consideration of neoadjuvant therapy prior to resection. Alternatively, a same-day streamlined approach adopted at The Chest Disease Center includes two integrated pathways: (I) mark and resect and (II) diagnose and resect if malignancy is confirmed, both performed within a single anesthesia episode. These pathways highlight the advantages of a coordinated, multidisciplinary workflow in optimizing lung cancer diagnosis and treatment. The figure was created on an online platform (Canva). The editable online version can be accessed at https://www.canva.com/design/DAGIbMbqQFw/dF4uvGDcFdapwGLcCuitrw/edit?utm_content=DAGIbMbqQFw&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton. IP, interventional pulmonology; MDC, multidisciplinary conference; Med Onc, medical oncology; Rad Onc, radiation oncology; TS, thoracic surgery.

complex non-infected and/or malignant pleural effusions.

PALs, defined as non-resolving pneumothorax with air leaks lasting more than 5–7 days, present with similar management challenges. Air leaks can prolong hospitalization, increase infection risk, and delay return

to baseline activity (53). Surgical repair is hailed as the gold standard, yet many patients, particularly those with significant comorbidities, are not fit for surgery. In such cases, bronchoscopic management led by IP may be considered. For instance, postsurgical air leaks can be

localized via bronchoscopy and subsequently treated via EBV deployment, sealants/autologous blood patch infusion, or pleural blood patching; these interventions are conducted by IP in close coordination with TS to determine optimal timing and necessity for therapeutic escalation (2,54).

Our pleural disease program at the Chest Disease Center integrates IP and TS into a shared decision-making framework. The majority of CPI and PAL cases are reviewed jointly, often at bedside, to select the best initial intervention and contingency plan. TS involvement from the outset ensures that surgical options remain on the table without delay, while IP provides minimally invasive strategies for those unsuitable for or unwilling to undergo surgery. We believe that this model creates a smooth escalation pathway, where patients are first offered the least invasive option, with both teams poised to move quickly to definitive surgical intervention if needed.

Conclusions

The Chest Disease Center at Beth Israel Deaconess Medical Center has established a collaborative MDT approach to manage patients, which is imperative to optimize patient-centered care. This has streamlined treatment for our patients, improving time-to-definitive care and overall outcomes. We have a highly skilled MDT of interventional pulmonologists and thoracic surgeons under one division to capitalize on the diagnostic precision and therapeutic efficacy of both specialties. The resulting synergy addresses the complexities of such diseases, which translates into optimized and comprehensive patient care. A unified division facilitates the referral process, allowing for careful patient selection for specific diagnostic and treatment modalities, and takes the timing factor into consideration. The Chest Disease Center includes specialized integrated programs, such as the Lung Cancer Program, Complex Airway Program, Pleural Disease Program, Benign Esophageal Disease Program, and COPD Clinic (55). This ensures that patients receive expert care tailored to their specific conditions. The team also actively and regularly collaborates with colleagues in general pulmonary medicine, critical care, medical oncology, radiation oncology, and otolaryngology to develop individualized care plans.

A key factor contributing to the Chest Disease Center's success is active communication between all team members. Both primary specialties (e.g., IP and TS) attend and play a crucial role in patient rounds, discuss cases, coordinate care, and hold educational curriculums for faculty and trainees

on a regular, consistent basis. This constant interaction nurtures a collaborative environment that enhances the quality of care. Furthermore, collaboration at the Chest Disease Center extends beyond clinical practice into robust research endeavors, as it is actively leading multiple highly relevant clinical trials and research projects aimed at pushing the standards of patient care even higher.

This successful model at The Chest Disease Center provides a blueprint for adoption in other hospitals and healthcare institutions. By fostering collaboration between interventional pulmonologists and thoracic surgeons, healthcare providers can deliver custom, high-quality care that addresses the advancing complexities of pulmonary diseases, improving patient satisfaction and QoL.

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Footnote

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